Research Interests and Proposals

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My research interests are broadly in macroeconomics and econometrics, particularly in expectations, heterogeneity and the link from micro to macro. I enjoy developing and refining models to understand how differences in agents' information, beliefs and economic circumstances shape their decisions and responses to policy. For instance, technological change such as the expansion of AI can create aggregate uncertainty that I am interested in examining through firms' expectations and investment commitments. I also explore how economic structure and prior knowledge can guide estimation and sharpen inference. In earlier work, I have used economic priors to guide high-dimensional variable selection and examined how pay composition shapes workers' perceptions of income risk.

The three proposals below develop these questions through simple models, preliminary evidence and empirical comparisons that I am working to establish and develop further during my PhD.

1. Reading the Missing Intercept p. 2

Cross-sectional comparisons can identify how responses differ across households or regions while leaving their common equilibrium response unobserved. I ask whether household decision rules and market clearing can recover part of this missing response. A toy model shows how known spending propensities and an observed consumption difference can identify income feedback. In a regional application to stock-wealth gains, I show how differences in the composition of earnings responses can separate direct wealth spending from feedback under common spending behaviour. I am working to establish a credible empirical comparison and the additional restrictions needed to connect local estimates to national effects.

2. Overbuilding on a Real Technology: Shale and AI p. 13

I study whether firms interpret favourable news too optimistically when committing capital, and how the resulting investment affects returns and repayment obligations when earnings disappoint. A model with diagnostic expectations separates the expected cost of initial overcommitment from the consequences of later news. The US shale expansion provides a setting in which to examine these decisions: preliminary evidence shows forecast reversals and persistent production and debt after spending contracts. I aim to connect expectations formed before investment to firms' commitments and subsequent returns, using shale to inform the same questions about the AI buildout.

3. AI Investment and the Economic Life of Chips p. 24

This proposal presents a model I wrote for an idea I am developing to present to Ricardo Reis, as part of our ongoing research on how to model capital depreciation in the age of AI. I study how demand growth, technological obsolescence and constraints on new-chip production jointly affect investment and the earnings and values of existing chips. The model distinguishes physical failure and declining capabilities from changes in the returns those capabilities earn: an older chip can earn more even as it falls behind newer equipment. Preliminary rental evidence motivates this distinction; measuring asset values and capabilities remains central to the empirical work.

Reading the Missing Intercept

1.1 The idea

The missing-intercept problem limits what cross-sectional estimates can tell us about aggregate effects (see Moll, 2025, for an exposition). For instance, suppose a programme transfers income from households with a low propensity to consume to households with a high propensity to consume. The recipients spend more of each additional euro than the households financing the programme, raising demand and generating earnings that support further consumption. A typical empirical approach would compare consumption changes across household groups while including time fixed effects to absorb aggregate movements. These fixed effects also absorb any response to the programme shared by all households, such as a common increase in consumption financed by the additional earnings it generates. This leaves part of its general-equilibrium effect unidentified even when the relative response is estimated correctly. But what if we could use the structure of an economic model to recover some of this feedback from the differences in household responses that we do observe? For instance, could information on households' spending propensities help us establish how much of their consumption response is financed by the additional earnings the programme generates?

Wolf (2023) tackles part of this problem by combining household consumption estimates with fiscal-multiplier evidence to recover aggregate responses. His demand-equivalence result establishes conditions under which private spending and government purchases generate the same equilibrium feedback. Applying it requires a fiscal comparison with matching demand paths, financing and macroeconomic conditions, alongside assumptions ensuring equivalent transmission. These requirements can be difficult to justify in a particular application, although Wolf also examines how departures from exact equivalence affect the recovery.

Manuel (2026) approaches the problem from an econometric perspective, which provides the inspiration for this proposal. He shows that households' heterogeneous responses to general-equilibrium effects can enter cross-sectional slopes, preventing them from identifying direct effects even when the aggregate shock is exogenous. His analysis makes explicit how identification depends on the comparison made across households and the controls included in the regression.

My idea is to combine these approaches by using the structure of an economic model to derive restrictions for econometric identification, for instance of the part of the consumption response financed by additional earnings. Starting from a specified model, such as a New Keynesian model with heterogeneous households, I would use household decision rules and market clearing to investigate whether differences in observed consumption and earnings responses allow us to identify how much direct spending and income feedback each contribute to the overall consumption response. For example, if we know households' spending propensities, each possible consumption response implies an income change needed to support it. Requiring these changes to be consistent with the income generated by additional spending may allow us to recover the common response without first identifying how the earnings are distributed. The aim is to establish which components of feedback we can recover and what assumptions are needed, including when suitable fiscal evidence is unavailable.

In this proposal, I first outline the idea through a simple toy model: two household types have known marginal propensities to consume (MPCs), but receive unknown shares of the income generated by a transfer programme. Their observed spending difference alone leaves the aggregate response undetermined. However, household spending rules and market clearing supply an additional restriction: the income needed to support their consumption must equal the income generated by their purchases. Combining this restriction with the observed spending difference recovers aggregate consumption and, after subtracting the direct spending response implied by the transfers and MPCs, identifies the contribution of income feedback.

To finish this proposal, I outline a pilot regional application based on Chodorow-Reich, Nenov and Simsek (2021), where they find that additional hours generate earnings for hand-to-mouth households, while higher wages also reach stockholders who save part of the gain. In a simplified adaptation of their model, I show how differences in wage adjustment across groups can identify direct wealth spending and income feedback without supplying a local multiplier, provided household spending behaviour is common and the propensity to spend extra hours income is known. Using their replication data, I find positive payroll and employment responses to stock-wealth gains, with payroll rising by more than employment. These results motivate examining earnings composition, although establishing a credible group comparison and translating the measured responses into the model remain empirical tasks that I am currently working on. The goal of continued research would be to estimate the local spending generated through income feedback per dollar of wealth gain, and to establish what additional restrictions would be required to connect this estimate to the national missing-intercept problem.

1.2 A toy model: recovering what a difference leaves out

Consider two equally sized household types. Type 1 spends a fraction b_1 of each additional dollar of disposable income and type 2 spends b_2, where $0 < b_2 < b_1 < 1$. A programme gives $\tau > 0$ to each type-1 household and taxes each type-2 household by the same amount. The programme raises initial demand because it transfers resources to households that spend a larger fraction of them, even though the transfer itself leaves average income unchanged. The aim is to identify how much of the programme's aggregate consumption response comes directly from the transfers and how much comes from the additional earnings generated by the resulting spending.

Let c_i be type i's consumption response and Y the resulting income increase per household. Type i receives $a_i Y$ of this additional income, where the unknown shares satisfy $a_i \geq 0$ and $(a_1 + a_2)/2 = 1$. Prices and interest rates are fixed, firms meet demand, and consumption is the only expenditure component that changes. Household spending and market clearing then give

$$c_1 = b_1(\tau + a_1 Y), \qquad c_2 = b_2(-\tau + a_2 Y), \qquad Y = \frac{c_1 + c_2}{2}. \tag{1.1}$$

The unknown income shares matter: directing more of the earnings gain to type 1 creates more spending in subsequent rounds. Knowing the transfers and MPCs therefore does not by itself determine the aggregate response.

Suppose a research design identifies the programme's causal spending difference, $c_1 - c_2$, but cannot identify a movement common to both types. Dividing each consumption response by its MPC recovers the disposable income change that supports it. Averaging across households gives

$$\frac{1}{2}\left(\frac{c_1}{b_1} + \frac{c_2}{b_2}\right) = \frac{1}{2}\left[\tau + a_1 Y - \tau + a_2 Y\right] = Y = \frac{c_1 + c_2}{2}. \tag{1.2}$$

The transfers cancel because their financing is balanced, and the income shares sum to all additional income. The left-hand side measures the income needed to support spending; the right-hand side measures the income created by that spending. This equality supplies the second restriction needed to recover the two responses from their observed difference.

Substituting $c_1 = Y + (c_1 - c_2)/2$ and $c_2 = Y - (c_1 - c_2)/2$ into (1.2) yields

$$Y = \frac{b_1 - b_2}{2(b_1 + b_2 - 2b_1 b_2)}(c_1 - c_2). \tag{1.3}$$

The denominator is positive because $b_1 + b_2 - 2b_1 b_2 = b_1(1 - b_2) + b_2(1 - b_1) > 0$. Adding the same amount to both consumption responses would preserve their observed difference, but would change the income needed to finance them by more than the income generated by their extra purchases, since both MPCs are below one. The model selects the common response consistent with both household spending and production. Subtracting initial demand, $(b_1 - b_2)\tau/2$, then gives spending due to income feedback.

$$\underbrace{\frac{b_1 a_1 + b_2 a_2}{2} Y}_{\text{income feedback}} = \underbrace{Y}_{\text{recovered total}} - \underbrace{\frac{(b_1 - b_2)\tau}{2}}_{\text{known direct response}}. \tag{1.4}$$

The example makes precise what the model contributes, but also what the data must identify. The MPCs must apply to both transfers and earnings, and $c_1 - c_2$ must include the programme's differential equilibrium effects. A household experiment measuring only the response to an individual's own transfer does not automatically identify that difference. The regional application uses payroll and labour responses to obtain information about the earnings generated by the initial spending increase.

1.3 A simple regional model of income feedback

The regional model I present below is a simple adaptation of Chodorow-Reich et al. (2021). I retain stockholders, hand-to-mouth households and local spending, while simplifying production and wage adjustment. In their paper, they separate direct wealth spending from income feedback using an externally calibrated local multiplier. I propose identifying the same decomposition from differences in regional wage and hours responses, using the model's restrictions on who receives additional earnings and how much they spend. The model describes a relative wealth experiment across small regions, holding national prices, interest rates and future earnings fixed.

1.3.1 Who receives the additional earnings?

I first describe how additional earnings are distributed across households and how this determines the spending they generate. A share $\theta \in (0,1)$ of each region's households lives hand to mouth and spends all current earnings. The remaining households hold stocks and spend a fraction $\rho \in (0,1)$ of additional wealth or current earnings. I assume the same spending propensity out of both sources; Appendix 1.A.1 gives a household problem that delivers this rule. Stocks are claims outside the region, so additional local earnings do not generate a further dividend payment.

Both types initially work $\bar{L}$ hours per household at wage $\bar{W}$. Following Chodorow-Reich, Nenov and Simsek (2021),[1] I assume that stockholders keep their hours fixed, while hand-to-mouth households accommodate the extra hours assigned by firms. Additional hours therefore generate income for households that spend all of it. Higher wages instead raise both types' earnings, including those of stockholders who save part of the gain. How payroll growth divides between hours and wages consequently matters for further spending.

Normalise the mass of households to one and let E denote total nominal household consumption expenditure in the region. Let x denote the wealth gain as a fraction of baseline payroll $\bar{W}\bar{L}$. The proportional changes in total hours and the wage are $\ell = dL/\bar{L}$ and $w = dW/\bar{W}$, respectively.

Hand-to-mouth households receive all additional hours income and a share θ of the income generated by higher wages. Their earnings gain, relative to baseline payroll, is therefore $\ell + \theta w$ to first order. Stockholders receive the remaining wage income, $(1-\theta)w$, alongside the initial wealth gain. Applying each type's spending propensity gives

$$\begin{aligned} \frac{dE}{\bar{W}\bar{L}} &= \underbrace{\ell + \theta w}_{\text{hand-to-mouth spending}} + \underbrace{\rho x + \rho(1-\theta)w}_{\text{stockholder spending}} \\ &= \underbrace{\rho x}_{\text{direct wealth spending}} + \underbrace{\ell + m_w w}_{\text{income feedback}}, \qquad m_w = \theta + (1-\theta)\rho. \end{aligned} \tag{1.5}$$

The coefficient on hours income is one because its recipients spend the whole gain. The coefficient $m_w \in (\rho, 1)$ is the average propensity to spend additional wage income, weighted by households' initial earnings. The object of interest is the income-feedback component, $\ell + m_w w$: additional spending financed by the local earnings response.

1.3.2 How does spending return as income?

I next describe how household spending generates local earnings, completing the feedback between income and consumption. Households spend a fixed share $\eta \in (0,1)$ on local nontradables and the remainder on tradables produced elsewhere. Competitive firms use labour alone, so nontradable payroll equals ηE. Outside buyers spend a fixed nominal amount on the region's tradable output, implying unit-elastic export demand and unchanged tradable payroll. At a balanced-trade benchmark, total payroll equals expenditure: $\bar{W}\bar{L} = \bar{E}$.

[1] See their Section IV.A and footnote 30, which explains how wage adjustment affects the distribution of additional earnings and hence the local multiplier.

Let n denote the proportional change in nontradable payroll. Since this payroll equals ηE, it grows at the same rate as household expenditure. Total payroll grows only through the nontradable sector, which initially accounts for a share η of total payroll. These conditions give, to first order,

$$n = \frac{dE}{\bar{E}} = \frac{dE}{\bar{W}\bar{L}}, \qquad \ell + w = \eta n. \tag{1.6}$$

Only the fraction η of additional spending therefore returns as local earnings; the remainder pays for imports. A wage rule $w = \kappa \ell$, with $\kappa \geq 0$, determines how payroll growth divides between wages and hours. Firms meet demand at the resulting wage, and hand-to-mouth households supply the assigned hours; I do not impose an additional labour-supply condition.

Together with the household spending rule in (1.5), these equations close the regional model. Successive rounds of spending and earnings generate a unique equilibrium, whose multiplier is derived in Appendix 1.A.2.

1.4 What do the responses identify?

I now show when regional payroll and hours responses identify direct wealth spending and income feedback under the model's restrictions. Suppose a research design identifies the causal responses of nontradable payroll, total payroll and total hours to the normalised wealth shock: β^N, β^P and β^L. Outcomes are proportional changes, and all three responses concern the same horizon and population. Since payroll growth is wage growth plus hours growth to first order, $\beta^W = \beta^P - \beta^L$ measures the wage response. Combining the spending rule in (1.5) with market clearing in (1.6), and expressing responses per unit of the wealth shock, gives

$$\underbrace{\beta^N}_{\text{total spending per wealth dollar}} = \underbrace{\rho}_{\text{direct spending}} + \underbrace{\beta^L + m_w \beta^W}_{\text{income feedback}}. \tag{1.7}$$

The normalisation makes the three terms comparable: β^N measures additional spending per dollar of wealth, while β^L and β^W measure additional earnings from hours and wages per dollar of wealth. The target is $\beta^N - \rho$, the amount of spending financed by those earnings. It is a nominal expenditure response; recovering real consumption would also require the price response.

With fixed wages, all additional earnings come from extra hours and reach hand-to-mouth households, who spend the entire gain. In that case $\beta^W = 0$ and $\beta^P = \beta^L$, so feedback equals the payroll response and $\rho = \beta^N - \beta^P$. When wages also rise, some earnings reach stockholders and a single regional response leaves two unknowns, ρ and m_w, in one equation.

To separate them, I compare two groups of regions $g \in \{1, 2\}$ with common household parameters ρ, θ and expenditure share η, but different wage-adjustment parameters κ_g. Both groups therefore have the same spending propensities. Their earnings gains can differ in composition. After subtracting spending financed by hours, define $Z_g = \beta^N_g - \beta^L_g$. The two groups supply

$$Z_1 = \rho + m_w \beta^W_1, \qquad Z_2 = \rho + m_w \beta^W_2. \tag{1.8}$$

Taking the difference between groups removes the common direct wealth effect. The remaining spending difference is financed by additional wage income, so dividing it by the difference in

wage-income responses identifies m_w. Subtracting this wage-financed spending from either group's Z_g then recovers ρ.

Proposition 1.1 (Identification from earnings composition)**.** *Under the regional model, identified causal responses in two groups with $\beta_1^W \neq \beta_2^W$ identify the direct MPC and income feedback:*

$$m_w = \frac{Z_2 - Z_1}{\beta_2^W - \beta_1^W}, \qquad \rho = Z_1 - m_w \beta_1^W, \qquad \textit{feedback in group } g = \beta_g^N - \rho. \tag{1.9}$$

Proof. Subtract the equations in (1.8) to recover m_w, then substitute into either equation to recover ρ. Equation (1.7) gives the remaining spending component. □

Different wage rules generate different wage responses in this model, as Appendix 1.A.2 verifies. Recovery requires common household spending behaviour and the assumption that recipients spend all additional hours income. If MPCs or household shares differ freely across groups, earnings composition no longer isolates the unknown propensities. A small wage-response difference also makes the calculation imprecise. The recovered values must satisfy $0 < \rho < m_w < 1$, and the model requires $\beta_g^P = \eta \beta_g^N$; these implications provide checks on the proposed interpretation.

1.4.1 Sensitivity to spending out of hours income

I next examine how identification depends on the assumption that households spend all additional hours income. Suppose the households receiving those hours instead spend a common fraction m of additional earnings, where $\rho < m \leq 1$. They are no longer strictly hand to mouth when $m < 1$, but the allocation of hours and wages is unchanged. The spending equation becomes

$$\beta_g^N = \rho + m\beta_g^L + m_w \beta_g^W, \qquad m_w = \theta m + (1 - \theta)\rho. \tag{1.10}$$

For a given m, subtracting $m\beta_g^L$ from each group's spending response again leaves two equations in ρ and m_w, which can be solved as in (1.9). I can therefore examine how the inferred direct MPC and feedback change across externally justified values of this propensity, retaining only values consistent with $0 < \rho < m_w < m$.

If m is also unknown, different spending propensities can reproduce the same observed responses. With a common local spending share η and an unknown household share θ, market clearing links hours and wages so that additional wage-adjustment groups cannot resolve this ambiguity. Appendix 1.A.3 shows the trade-off: among admissible values consistent with the same responses, a lower m implies a larger direct wealth effect and less income feedback. The empirical exercise must therefore either justify the hours-income propensity or report the decomposition's sensitivity to it.

1.5 An empirical illustration and the next comparison

Chodorow-Reich et al. (2021) estimate local labour-market responses to stock-wealth gains and use external evidence on the local multiplier and labour share to recover a direct wealth MPC. Using their replication data, I reproduce the payroll and employment responses and examine what these outcomes can tell us about the income feedback described above.

1.5.1 What is estimated?

For county a, shock quarter t and horizon h, let $q^k_{a,t+h}$ be the log change in outcome k from $t-1$ to $t+h$. I estimate

$$q^k_{a,t+h} = \alpha^k_{a,h} + \delta^k_{\text{state}(a),t,h} + \beta^k_h x_{at} + C'_{at,h}\gamma^k_h + u^k_{a,t+h}. \tag{1.11}$$

The wealth shifter x_{at} is lagged county stock wealth multiplied by the portfolio-adjusted stock return, divided by annual IRS wage income. County effects absorb persistent local differences and state-by-quarter effects absorb movements shared within a state. I retain the authors' controls and population weights, pooling shock dates from 1992Q3 to 2015Q2. Interpreting the estimates as local wealth responses requires conditional wealth exposure not to predict responses to other shocks or differential responses to the national consequences of the stock-market shock itself (Manuel, 2026).

Table 1.1 reports the responses at the two-year horizon. These measure changes from $t-1$ to $t+7$, rather than cumulative earnings over those quarters.

Table 1.1: Full-sample responses to stock-wealth gains

Outcome	Estimated response
Nontradable payroll	3.24 (1.01)
Total payroll	2.19 (0.63)
Total employment	0.78 (0.36)
Average earnings per worker	1.41 (0.49)

Notes: Responses in basis points to a stock-wealth gain equal to 1% of annual county IRS wage income, measured from $t-1$ to $t+7$. Standard errors clustered by county and quarter appear in parentheses. All outcomes use the same sample of 2,896 counties and 263,210 county-quarter observations, controls and population weights.

1.5.2 What does the evidence contribute?

The positive nontradable and total payroll responses show that stock-wealth gains generate additional local earnings. Figure 1.1 shows how these responses develop over time: payroll and employment increase over the quarters following the wealth gain, and the earnings increase persists beyond the initial quarter. Earnings per worker also rise, so employment growth accounts for only part of the payroll response.

Because the outcomes use the same sample and specification, the payroll response decomposes exactly into employment and average earnings per worker: $\log P = \log J + \log(P/J)$, where P is total payroll and J is employment. At the two-year horizon, 0.78 of the 2.19 basis points comes from employment and the remaining 1.41 from earnings per worker. The latter includes hours per worker, wages and workforce composition, so it cannot yet be interpreted as the model's wage response.

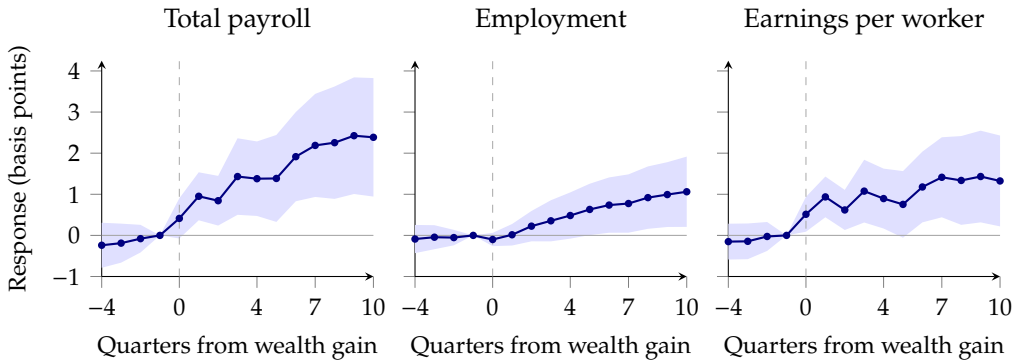


Figure 1.1: Payroll, employment and earnings per worker over time

Notes: Responses in basis points to a wealth gain equal to 1% of annual county IRS wage income. Outcomes are log changes relative to $t-1$, whose response is zero by construction. Shading denotes pointwise 95% confidence intervals, with county-and-quarter clustered standard errors. Each horizon uses the same sample across outcomes; observations vary slightly across horizons because of data availability. Earnings per worker includes hours, wage and workforce-composition changes.

I have not yet worked through the comparison between groups fully. Building on the pooled estimates above, I outline the regressions I would run to apply the model's identification argument. For each outcome k, comprising nontradable payroll, total payroll and employment, I would estimate

$$q^k_{a,t+h} = \alpha^k_{a,h} + \delta^k_{\text{state}(a),t,h} + \beta^k_{1,h} x_{at} + \lambda^k_h D_a x_{at} + C'_{at,h}\gamma^k_h + u^k_{a,t+h}. \tag{1.12}$$

The indicator D_a equals one for counties in the second group, so the estimated responses are $\beta^k_{1,h}$ and $\beta^k_{1,h} + \lambda^k_h$. I would use the same observations across outcomes and retain the existing controls, population weights and standard errors clustered by county and quarter.

These estimates provide the group responses needed for the decomposition in Section 1.4. In the simplified model, subtracting spending financed by additional hours leaves direct wealth spending and spending out of higher wages:

$$Z_g = \beta^N_g - \beta^L_g = \rho + m_w \beta^W_g.$$

Taking the difference between groups removes the common direct wealth effect ρ. Dividing the remaining spending difference by the difference in wage-income responses therefore identifies m_w. Subtracting $m_w\beta^W_g$ from either group's Z_g then recovers ρ, and total income feedback equals $\beta^N_g - \rho$. This recovery assumes common household spending propensities across groups. Applying it to the data requires the hours, workforce-composition and labour-share adjustments discussed in Appendix 1.A.5.

1.6 Preliminary conclusion

I conclude by summarising what this work in progress establishes and outlining the next steps. The toy model shows how household spending rules and market clearing can identify income feedback: the transfers and known MPCs determine direct spending, while the observed consumption difference and the equilibrium restriction recover total consumption. The regional model considers a setting in which the direct wealth MPC is unknown. Under its assumptions, differences in wage

and hours responses across regions with common spending behaviour provide the additional information needed to separate direct spending from income feedback.

The next step is to establish whether the available data support such a comparison, particularly whether regions with different wage adjustment have sufficiently comparable household spending behaviour. If the comparison supports the model's restrictions, I would estimate direct spending and income feedback, accounting for the measurement issues discussed above and reporting uncertainty and sensitivity to the assumptions needed for recovery. This would allow me to assess whether regional responses can identify the spending financed by additional earnings without supplying an external local multiplier.

References

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1.A Supporting derivations and measurement

1.A.1 A household foundation for the spending rule

A stockholder chooses current and future nominal expenditure, taking prices P_0, P_1, the gross interest rate R and lifetime resources V^s as given:

$$\max \ \rho \log(E_0^s/P_0) + (1-\rho)\log(E_1^s/P_1) \quad \text{subject to} \quad E_0^s + E_1^s/R = V^s. \tag{1.A.1}$$

The first-order condition gives $E_1^s/R = (1-\rho)E_0^s/\rho$, hence $E_0^s = \rho V^s$: an extra dollar of wealth or current earnings raises spending by ρ when future earnings are fixed. Hand-to-mouth households spend current earnings and receive all additional hours: $L = \theta L^h + (1-\theta)\bar{L}$ implies $dL = \theta dL^h$. Differentiating earnings gives the allocation in (1.5). The rule $W = \bar{W}(L/\bar{L})^\kappa$ implements wage adjustment, with firms assigning hours and households accommodating them.

1.A.2 Equilibrium and the identifying wage difference

In the household toy, substituting the spending rules into market clearing gives $Y = (b_1 - b_2)\tau/(2 - b_1 a_1 - b_2 a_2)$. The denominator is positive because $a_1 + a_2 = 2$ and both MPCs are below one, so the equilibrium used in Section 1.2 exists uniquely for every admissible income allocation.

In the regional model, payroll accounting and the wage rule give $\ell = \eta n/(1+\kappa)$ and $w = \kappa\eta n/(1+\kappa)$. Substitution into spending yields

$$n = \rho \mathcal{M} x, \qquad \mathcal{M} = \left[1 - \eta \frac{1+\kappa m_w}{1+\kappa}\right]^{-1}. \tag{1.A.2}$$

The feedback coefficient is positive and at most $\eta < 1$, so the first-order equilibrium is unique. Greater wage adjustment reduces amplification by directing more income to stockholders who save part of it. In this model the target $\beta_g^N - \rho$ equals $\rho(\mathcal{M}_g - 1)$.

The wage response and its derivative are

$$\beta_g^W = \frac{\eta\rho\kappa_g}{(1-\eta) + \kappa_g(1-\eta m_w)}, \qquad \frac{\partial \beta_g^W}{\partial \kappa_g} = \frac{\eta\rho(1-\eta)}{[(1-\eta) + \kappa_g(1-\eta m_w)]^2} > 0. \tag{1.A.3}$$

Thus different κ_g, with common household parameters and spending shares, generate the wage-response contrast required for identification.

1.A.3 What changes when hours recipients save?

Let the recipients of additional hours spend fraction m of current earnings, with $\rho < m \leq 1$. The multiplier becomes $[1 - \eta(m + \kappa m_w)/(1+\kappa)]^{-1}$, which remains positive and finite because $0 < m_w < m \leq 1$ and $\eta < 1$. Using $\beta_g^L = \eta\beta_g^N - \beta_g^W$ in (1.10) gives

$$\beta_g^N = A + B\beta_g^W, \qquad A = \frac{\rho}{1 - m\eta}, \qquad B = \frac{m_w - m}{1 - m\eta}. \tag{1.A.4}$$

Distinct wage responses identify the intercept A and slope B; market clearing identifies $\eta = \beta_g^P/\beta_g^N$. For any candidate m, the same responses are consistent with

$$\rho(m) = (1 - m\eta)A, \qquad m_w(m) = m + (1 - m\eta)B, \qquad \theta(m) = \frac{m_w(m) - \rho(m)}{m - \rho(m)}, \tag{1.A.5}$$

provided $0 < \rho(m) < m_w(m) < m \leq 1$. These inequalities ensure an admissible household share and a unique equilibrium reproducing the responses with unchanged wage rules. Around an interior solution, several m satisfy them; further groups with the same η trace the same line. Since $A > 0$, lowering m raises

$\rho(m)$ and lowers feedback $\beta_g^N - \rho(m)$. Differences in feedback still equal $\beta_2^N - \beta_1^N$ under a common direct MPC.

1.A.4 Estimating the group comparison

For two groups, the regression in (1.11) replaces $\beta_h^k x_{at}$ with $\sum_{g=1}^{2} \beta_{g,h}^k \mathbf{1}\{g(a) = g\} x_{at}$, allowing the wealth response to differ across them. A predetermined grouping alone does not justify common MPCs: characteristics affecting wage adjustment may also reflect household resources and stock ownership. I would assess these differences alongside the earnings responses, with controls allowing for differential exposure to national developments.

Before calculating a decomposition, I would estimate the group responses jointly, check the precision of the wage-response contrast and assess the model's common spending-share restriction. Employment must be translated into hours, and workforce composition accounted for before residual payroll growth can measure wage growth. Allowing capital also requires a labour share to translate nontradable payroll into spending. Appendix 1.A.5 explains these adjustments and the additional timing assumptions needed to connect a two-year response to the model.

1.A.5 Hours, capital and empirical normalisation

With an independently justified hours-to-employment response ratio χ_g, total hours and wages would be constructed as

$$\beta_g^L = \chi_g \beta_g^{\text{emp}}, \qquad \beta_g^W = \beta_g^P - \chi_g \beta_g^{\text{emp}}. \tag{1.A.6}$$

The original paper uses $\chi = 1.5$. Applying it across groups assumes comparable hours-per-worker adjustment; composition changes must also be accounted for before interpreting the residual as a common wage. Table 1.1 imposes no value of χ.

With constant sectoral labour shares s_N, s_T, nontradable payroll is $s_N \eta E$. At the balanced-trade benchmark, total payroll is $\bar{s}\bar{E}$, where $\bar{s} = \eta s_N + (1 - \eta) s_T$. Nontradable payroll still changes proportionally with expenditure, so

$$\frac{\beta_g^N}{\bar{s}} = \rho + \beta_g^L + m_w \beta_g^W, \qquad \text{feedback in group } g = \frac{\beta_g^N}{\bar{s}} - \rho. \tag{1.A.7}$$

The two-equation argument survives conditional on the labour share and the household-income assumptions. This adjustment does not implement the original paper's full national model.

The regressions use annual IRS wage income and quarterly payroll growth. A structural interpretation requires consistent flow units and income populations, and measured nontradable industries that represent the model's sector. The $h = 7$ estimate is an endpoint response, not cumulative spending. Mapping it into the model's single period additionally requires assumptions about intervening saving and responses to expected future income; matching units alone does not justify the static decomposition.

Figure 1.1 retains the horizon-specific industry control and uses common observations across outcomes at each horizon. Regressing log payroll minus log employment directly incorporates their covariance in the earnings-per-worker standard error. Pointwise intervals use the county-and-quarter clustered standard errors and corresponding t critical value; the $h = -1$ response is zero by construction.

Overbuilding on a Real Technology: Shale and AI

2.1 Introduction

The debate over a possible AI bubble raises a question about how firms invest when a new technology creates substantial but uncertain prospects of profitability. I study whether firms interpret favourable developments too optimistically when committing capital, and how the additional investment affects their returns and financial exposure when earnings disappoint. The AI buildout makes these decisions consequential: Van Nieuwerburgh (2026) estimates that a representative 200 MW training campus requires approximately $8.2 billion in capital expenditure and documents increasing reliance on leases, project debt and other external financing as spending outgrows internal cash flows. Firms must therefore assess future earnings before committing resources whose costs may remain payable if demand falls short.

To judge whether these commitments are excessive, I compare firms' choices with an explicit benchmark for investment. For example, Rungcharoenkitkul (2026) studies firms competing for dominant positions in AI, where the incentive to capture future business induces investment beyond the socially efficient level and financing allows firms to commit early. Caballero (2026) shows that temporary optimism can instead leave a lasting productive expansion when it supports enough investment to reach a self-sustaining high-capital allocation. These results distinguish the productive consequences of an expansion from the sustainability of valuations and the returns earned by investors. At the firm level, I focus on a firm's decision under uncertainty, comparing its investment with the programme that would maximise expected project value under the same information and technology.

I intend to use diagnostic expectations as a mechanism linking favourable news to that investment comparison. In Bordalo et al. (2018), agents overweight outcomes made relatively more likely by recent information, so an improvement in prospects can also make high returns seem too likely. Bordalo et al. (2026) connect this overreaction to investment and borrowing during expansions and to subsequent reversals. I use a simple version of this mechanism to ask which commitments reflect improved prospects and which reflect an excessively optimistic assessment of them. Empirically, I aim to connect expectations formed before spending to development expenditure, operating returns and financing outcomes in the same expansion.

The US shale boom provides an earlier setting in which those decisions and their consequences can be observed. Horizontal drilling and hydraulic fracturing made previously difficult oil and gas deposits accessible, while high oil prices encouraged development (Kleinberg and Paltsev, n.d.). The sharp price decline from mid-2014, associated with both supply and demand developments, then reduced the earnings available from installed capacity (Baumeister and Kilian, 2016). Supporting my narrative, Figure 2.1 follows Continental Resources and SM Energy over the expansion and its aftermath. After 2014, expenditure contracted sharply while output initially rose and debt remained elevated. This divergence motivates the distinction between continued production and the returns earned on earlier commitments, since producing oil after a price decline need not imply that the original investment recovered its costs. Shale therefore offers a setting in which to examine whether expectations during an expansion were overly optimistic and whether those assessments

preceded larger commitments, while distinguishing the initial decision from adverse news that arrived afterwards.

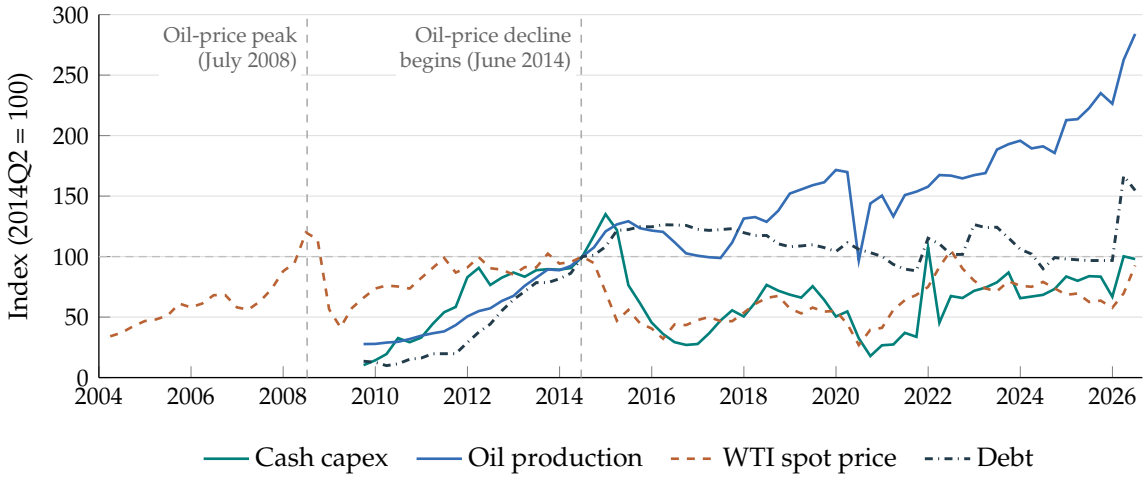


Figure 2.1: Continental Resources and SM Energy: full history

Notes: Quarterly totals for both firms, 2009Q3–2026Q2; 2014Q2 = 100. Capex is nominal oil and gas investment; production is crude oil/condensate; debt is measured at quarter-end. WTI starts in 2004Q1. Sources: Compustat, SEC filings and EIA. Construction details: Appendix 2.B.2.

To study this behaviour, I develop a two-period model in which a firm receives a signal about operating profitability and chooses an irreversible capital programme with increasing marginal installation costs. The Bayesian firm expands after favourable news; a diagnostic firm assigns a higher value to additional capital and expands further. Comparing the two programmes separates the expected cost of overcommitment from the effect of subsequent news on their realised returns. The additional capital can earn positive operating income while failing to recover its installation expenditure. When debt finances a fixed share of expenditure at unchanged terms, the larger programme also requires higher earnings per unit to meet promised repayment.

The preliminary shale evidence points to forecast reversals and shows that increases in earnings expectations accompany higher forecasts of capital expenditure, although the link to actual spending remains unresolved. I aim to connect these expectations to firms' investment decisions and subsequent returns, using the model to distinguish commitments justified by improved prospects from those induced by optimism. The broader goal is to understand how firms can overcommit to a productive technology and how the resulting exposure affects shareholders and creditors when earnings disappoint, using shale to inform the same questions about the AI buildout.

2.2 The firm, information and capital commitments

2.2.1 Investment before earnings are known

A risk-neutral firm chooses how much capital to install before knowing what the programme will earn. At date 0 it receives information about profitability, selects $k \geq 0$ and pays

$$C(k) = \frac{c}{2}k^2, \qquad c > 0. \tag{2.1}$$

The firm is already active, and expanding its programme requires increasingly expensive sites, equipment or organisational resources. The parameter c determines how quickly marginal installation cost, ck, rises, and $C(k)$ measures the firm's total installation expenditure.

At date 1 the programme generates Ak, where A is earnings per unit net of operating costs and could be negative. This payment summarises operating returns over the programme's life. Installation expenditure is sunk, capital has no separate resale value, and discounting between the two dates is normalised to one. Profitability per unit is taken as given, so the comparison across beliefs holds prices and technology fixed. The firm maximises expected project value, the expected operating earnings less installation expenditure. Funding is available; Section 2.4.2 subsequently introduces repayment obligations without changing this investment objective.

2.2.2 Information and its interpretation

Before receiving news, the firm believes that A is normally distributed with mean $\bar{A}$. It observes $s = A + \varepsilon$, where ε is independent normal noise with mean zero. Bayesian updating gives the conditional mean $\mu = \mathbb{E}[A \mid s]$. A signal above $\bar{A}$ raises μ, as a strong initial oilwell performance might improve the prospects for further development while leaving eventual earnings uncertain.

Following the Gaussian diagnostic-beliefs framework of Bordalo et al. (2019), the firm overweights profitability outcomes that become relatively more likely after the signal. I compare the Bayesian posterior with the posterior following a signal with no surprise, $s = \bar{A}$. These distributions have the same variance, so diagnostic reweighting shifts expected profitability to

$$m_\theta = \mu + \theta(\mu - \bar{A}), \qquad \theta \geq 0. \tag{2.2}$$

Appendix 2.A.1 derives this expression from the signal and the reweighted density. At $\theta = 0$, beliefs are Bayesian. With $\theta > 0$, favourable news raises expected earnings above the Bayesian forecast by an amount proportional to how much that forecast exceeds what the firm expected before receiving the news.

I compare programmes facing the same signal, installation costs and realised profitability. Following the subjective q-theory interpretation in Gennaioli et al. (2016), the marginal value assigned to capital is its expected operating return, which is μ under Bayesian beliefs and m_θ under diagnostic beliefs. I focus on favourable news with $\mu > \bar{A}$ and $\mu > 0$, so both valuations are positive. The following comparisons therefore concern how the interpretation of improved prospects changes a positive investment programme.

2.3 From beliefs to capital commitments

2.3.1 What would overreaction look like in expectations?

Diagnostic beliefs imply that an optimistic assessment can predict its own disappointment. Conditional on the signal, the true mean of profitability is μ, giving

$$\mathbb{E}[A - m_\theta \mid s] = -\theta(\mu - \bar{A}). \tag{2.3}$$

After favourable news, the expected forecast error is negative, meaning that the firm expects higher earnings than the available information justifies. Building on the forecast-predictability approach of Coibion and Gorodnichenko (2015) and Bordalo et al. (2020), I examine whether increases in earnings expectations predict subsequent forecast reversals as a preliminary check for corrections to earlier optimism.

Using I/B/E/S, I divide analysts covering the same firm's annual earnings target into two disjoint groups. I relate the earlier annual revision of group A to the subsequent annual revision of group B, pooling both group orientations:

$$r_{fth}^{B,\text{next}} = \alpha_t + \gamma_h + \beta r_{fth}^{A,\text{prior}} + u_{fth}. \tag{2.4}$$

Revisions are twelve-month changes in the log of each group's mean forecast; α_t and γ_h absorb target-year and horizon effects. Using disjoint groups prevents the same analyst's forecast from entering both revisions.

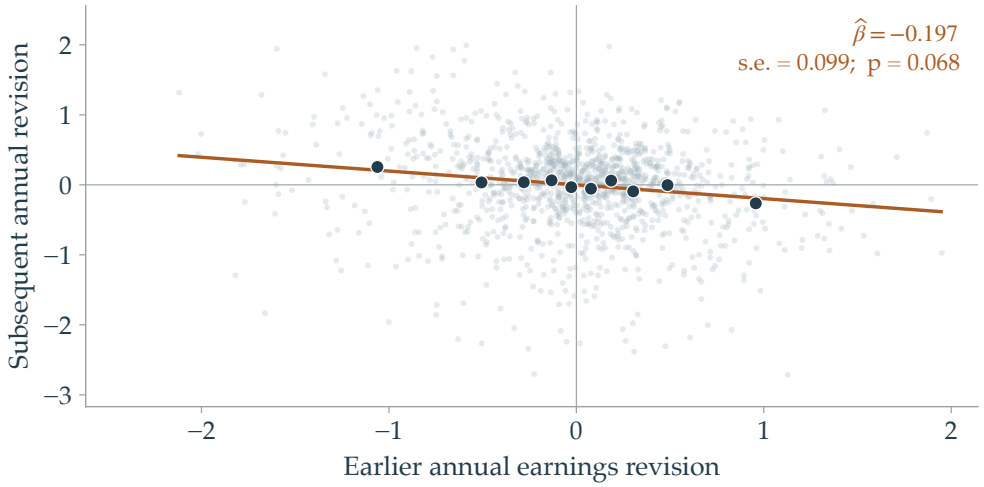


Figure 2.2: Increases in earnings expectations and subsequent forecast revisions

Notes: I/B/E/S: 643 target–horizon pairs, 103 firms, 2007–2021. Log revisions after 1% trimming and year/horizon adjustment. Dark dots: ten equal-count bin means; line: OLS on observations. Firm/year-clustered inference; see Appendix 2.B.1.

In Figure 2.2, larger earlier increases in earnings expectations predict lower subsequent revisions across disjoint analyst groups, providing preliminary evidence of forecast reversals. Controlling for target-year effects leaves the relationship largely unchanged, although the estimate is less precise. My next step is to compare the same targets and information dates using measures less sensitive to forecast noise, then connect expectations formed before spending to investment decisions.

2.3.2 How do beliefs change investment?

The firm invests until its perceived marginal return equals marginal installation cost. This connects the assessment of future earnings to capital commitments, as in the expectations-based investment framework of Bellifemine et al. (2026). Under the positive valuations considered here,

$$k_\theta = \arg\max_{k \geq 0} \left\{ m_\theta k - \frac{c}{2} k^2 \right\} = \frac{m_\theta}{c}, \qquad k_0 = \frac{\mu}{c}. \tag{2.5}$$

Both choices respond to improved prospects. Their difference measures the additional capital attributable to the diagnostic interpretation of the same news.

Proposition 2.1 (Optimism and capital commitments)**.** *Following favourable news with $\mu > \bar{A}$ and $\mu > 0$, a firm with $\theta > 0$ installs additional capital*

$$k_\theta - k_0 = \frac{\theta(\mu - \bar{A})}{c} > 0. \tag{2.6}$$

Its expenditure $C(k_\theta)$ also exceeds the Bayesian expenditure. Both differences increase with θ, holding information and technology fixed.

Proof. See Appendix 2.A.2.

Optimism raises the value assigned to the next unit and leads the firm further along its marginal cost schedule. A flatter schedule produces a larger capital response to the same belief distortion, while convex installation costs make expenditure rise more than proportionally with programme size. The empirical question is whether optimistic assessments formed before spending are associated with these additional commitments.

In the shale panel, I show that a one-standard-deviation increase in earnings expectations accompanies a rise in forecast capital expenditure of 0.78 percentage points of predetermined assets. I do not yet have causal evidence linking earnings expectations to realised investment, but a Granger test using oil prices does show that a ten-log-point increase in oil-price growth predicts 6.2 log points more capital-expenditure growth in the following quarter (see Appendix 2.B.3). Following Gennaioli et al. (2016), I aim to examine this link directly using managers' earnings expectations, investment plans and realised spending, asking whether overly optimistic forecasts lead to additional capital commitments.

2.4 Returns and financing consequences

2.4.1 Expected value and realised returns

Once capital is installed, its financial return depends on the profitability that materialises. For a fixed capital programme k, let $\Pi(k; A)$ denote realised project value and $V_0(k)$ its expected value conditional on the signal s, evaluated under Bayesian beliefs:

$$\Pi(k; A) = Ak - \frac{c}{2}k^2, \qquad V_0(k) = \mathbb{E}[\Pi(k; A) \mid s] = \mu k - \frac{c}{2}k^2. \tag{2.7}$$

The comparison with the Bayesian programme separates the cost of the initial decision from the subsequent news. Defining $L(A) = \Pi(k_0; A) - \Pi(k_\theta; A)$ and using $ck_0 = \mu$ gives

$$L(A) = \underbrace{\frac{c}{2}(k_\theta - k_0)^2}_{\text{expected cost of overcommitment}} - \underbrace{(A - \mu)(k_\theta - k_0)}_{\text{effect of subsequent news}}. \tag{2.8}$$

Appendix 2.A.3 supplies the intermediate algebra. The first term is positive because additional installation costs exceed the expected earnings of the units installed beyond k_0. The second term

records what those extra units earn when profitability differs from its conditional mean. Since $\mathbb{E}[A - \mu \mid s] = 0$, the true expected loss is

$$\mathbb{E}[L(A) \mid s] = \frac{c}{2}(k_\theta - k_0)^2 = \frac{\theta^2(\mu - \bar{A})^2}{2c}. \tag{2.9}$$

A decision that lowers expected project value can nevertheless yield a higher realised return. The larger programme performs better when $A > \mu + c(k_\theta - k_0)/2$, because sufficiently favourable news raises earnings on the extra units enough to cover their installation costs. Conversely, disappointing profitability increases the realised loss even when part of that loss reflects news unavailable at the investment date. The relevant assessment of overcommitment is therefore made using information available when capital is chosen, while the realised comparison measures how that choice and subsequent news jointly affect investors.

The programme can also remain productive while failing to recover its installation bill. For $k_\theta > 0$, positive net operating earnings and negative project value coexist when $0 < A < ck_\theta/2$. Increasing capital raises exposure to changes in A because each additional unit earns the same realised profitability, but neither positive production nor a subsequent earnings decline establishes whether the original programme was excessive. Project value here is evaluated at given prices and excludes consumer gains, spillovers and aggregate effects.

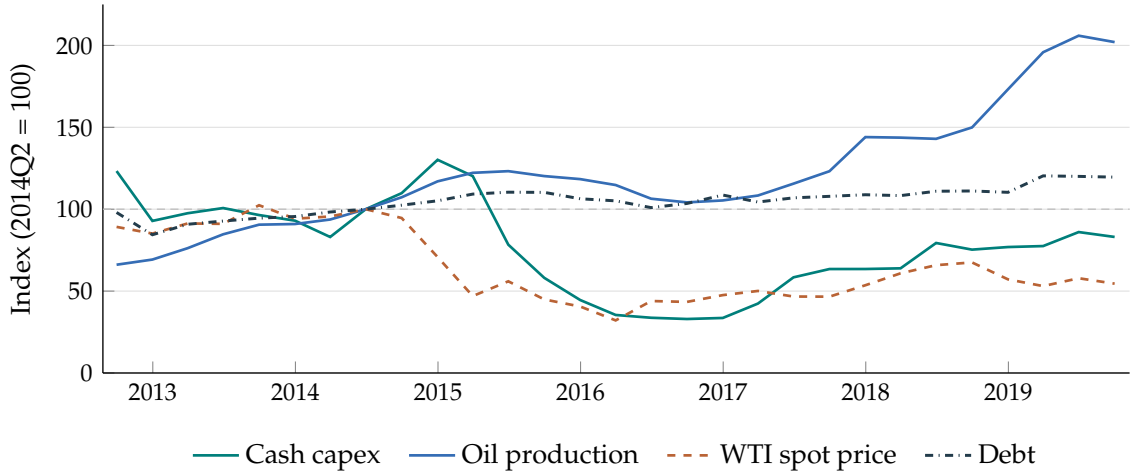


Figure 2.3: Investment contracts while production and debt persist

Notes: Quarterly totals for six producers, 2012Q3–2019Q3; 2014Q2 = 100. Capex is nominal oil and gas investment; production is crude oil/condensate; debt is measured at quarter-end. Sources: Compustat, SEC filings and EIA. Sample and construction: Appendix 2.B.2.

The shale evidence illustrates why production and investment returns need to be distinguished. Figure 2.3 follows six producers through the downturn: between 2014Q2 and 2015Q4, cash capital expenditure fell by 55.5 per cent, oil production rose by 18.4 per cent and the WTI oil price fell by 59.4 per cent. Firms curtailed new spending while earlier investments continued to support output. Anderson et al. (2018) show why output from existing wells can respond less to prices than drilling does, so this persistence need not imply an initially excessive programme.

I will link plans and forecasts made before investment to the subsequent operating cash flows of the same development programmes, accounting for asset transfers and commodity hedges. This would help assess whether earnings recovered investment costs and, using the model, compare the

observed programme with the programme implied by benchmark beliefs under the same realised profitability. Constructing that benchmark will also require an earnings assessment based on information available before investment and an estimate of the installation cost schedule.

2.4.2 Repayment commitments

I want to add borrowing to the model, motivated by its role in financing the AI buildout and the similar patterns I find in shale, to examine how firms commit to repayments before knowing the earnings available to meet them. A firm can reduce new expenditure after a downturn, but paying down earlier debt requires funds and reducing creditors' claims requires agreement or restructuring. To examine this exposure, suppose creditors supply a fixed fraction $\lambda \in (0, 1]$ of installation expenditure, $D_\theta = \lambda C(k_\theta)$, and are promised RD_θ at date 1, where $R > 0$ is fixed. Shareholders supply the remainder and the firm continues to maximise total project value. The financing assumptions describe repayment exposure at given terms, without introducing a separate equity-value objective or endogenous credit pricing, and the benchmark rules out subsequent funding or debt relief.

Proposition 2.2 (Capital commitments and repayment requirements)**.** *For a positive programme, earnings fall short of promised repayment if and only if*

$$A < a_\theta \equiv \frac{RD_\theta}{k_\theta} = \frac{R\lambda c}{2} k_\theta. \tag{2.10}$$

Following favourable news and with $\theta > 0$, the diagnostic programme has a higher repayment threshold:

$$a_\theta - a_0 = \frac{R\lambda c}{2}(k_\theta - k_0) = \frac{R\lambda\theta}{2}(\mu - \bar{A}) > 0. \tag{2.11}$$

Holding the true conditional distribution of profitability fixed, the probability of a repayment shortfall increases.

Proof. See Appendix 2.A.4.

The mechanism depends on expenditure per unit increasing with programme size. At a fixed debt-financing share, borrowing grows faster than capital, so the larger programme must earn more per unit to meet its obligations. When $0 < A < a_\theta$, it generates positive operating earnings but cannot cover promised repayment from those earnings alone. This implies default for a stand-alone project without other resources, while a firm may cover the shortfall with cash, other assets or new funding. A repayment shortfall reallocates losses between claimants and is not itself an additional resource loss to add to equation (2.8).

Descriptive shale evidence motivates examining this exposure. In a broader extraction sample, capital expenditure fell by 65 per cent between 2014 and 2016, while long-term debt fell by 20 per cent. Among 160 producers ranked by their 2013 leverage, recorded bankruptcy rates over 2015–2020 were 13 per cent in the lowest third and 26 per cent in the highest. Appendix 2.B.2 reports the underlying comparisons. Debt persistence and the leverage gradient describe exposure during the downturn, but do not identify the financing share of a particular programme or establish that the borrowing arose from excessive optimism.

Building on the relationship between drilling and debt studied by Gilje et al. (2020), I will link borrowing undertaken during expansion to repayment schedules, refinancing and restructuring

outcomes. I will use firms' 10-K and 10-Q filings, together with restructuring records, to trace how operating shortfalls translate into losses for shareholders and creditors. Having access to debt terms, repayments and recoveries might help establish who bears project losses, which could help me better understand the financial consequences of overinvestment.

2.5 Research programme and implications for AI

The model shows how overly optimistic earnings expectations can raise investment and repayment commitments, while the shale evidence provides preliminary signs of forecast reversals and persistent production and debt. I now want to connect these pieces by testing whether optimism formed before spending leads to larger investment programmes, then tracing their realised returns and the losses borne by shareholders and creditors. This requires matching forecasts and investment plans to the same programmes, with the longer-term aim of examining whether similar mechanisms operate in the AI buildout.

I will start with dated capital budgets, production guidance and earnings forecasts, and follow the same assets into operating cash flows. These data would help assess whether earnings recovered investment costs and compare the observed programme with the programme implied by benchmark beliefs. Debt terms, repayments and recoveries could then help establish who bears any losses. Beyond these comparisons, I want to extend the model to examine how capacity installed during an expansion affects prices and investment after optimism fades.

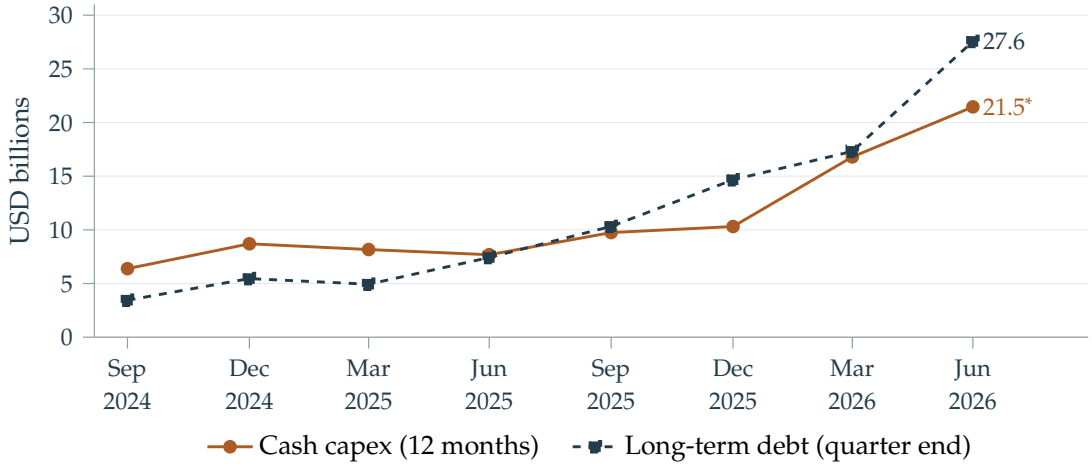


Figure 2.4: Capital expenditure and long-term debt at CoreWeave

Notes: Debt excludes current maturities and leases. *Uses revised 2025 cash capex; conflicting later filings imply $20.6bn. Sources: CoreWeave filings and earnings releases.

These questions also arise in the AI buildout. CoreWeave, a provider of computing capacity for AI, has expanded capital expenditure alongside substantial borrowing (Figure 2.4). Some of its borrowing is secured by equipment and contracted customer payments (CoreWeave, 2026), so assessing these commitments requires examining both the earnings anticipated when capacity is secured and the contracts supporting repayment. Applying the shale comparison therefore requires AI-specific expectations, costs and financing terms, including equipment depreciation, resale and leases. The broader goal is to distinguish expansion supported by improved prospects from additional commitments induced by optimism, and to trace how disappointing earnings affect shareholders and creditors even when the installed capacity remains useful. But while the AI buildout is still under way, can we really tell where justified expansion ends and excessive optimism begins, and who will be left with the losses if the expected earnings never arrive?

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2.A Derivations

2.A.1 Diagnostic beliefs

For independent $A \sim N(\bar{A}, \tau^2)$ and $\varepsilon \sim N(0, \sigma_\varepsilon^2)$ with positive variances, $A \mid s \sim N(\mu, v)$, where

$$\mu = \bar{A} + \frac{\tau^2}{\tau^2 + \sigma_\varepsilon^2}(s - \bar{A}), \qquad v = \frac{\tau^2 \sigma_\varepsilon^2}{\tau^2 + \sigma_\varepsilon^2}. \tag{2.A.1}$$

The reference posterior is $A \mid s = \bar{A} \sim N(\bar{A}, v)$. Diagnostic reweighting gives (Bordalo et al., 2019, Section III.B)

$$f_\theta(a \mid s) \propto f(a \mid s) \left[\frac{f(a \mid s)}{f(a \mid s = \bar{A})} \right]^\theta \propto \exp\left[-\frac{(a - m_\theta)^2}{2v} \right], \tag{2.A.2}$$

where completing the square gives $m_\theta = \mu + \theta(\mu - \bar{A})$.

2.A.2 Proof of Proposition 2.1

Strict concavity and the first-order condition $m_\theta - ck = 0$ give $k_\theta = m_\theta / c$ and $k_0 = \mu / c$. Hence $k_\theta - k_0 = \theta(\mu - \bar{A})/c > 0$ after favourable news. Since $C'(k) > 0$ for $k > 0$, expenditure also rises with θ. □

2.A.3 Project values

With $\Delta k = k_\theta - k_0$ and $ck_0 = \mu$,

$$C(k_\theta) - C(k_0) = \mu \Delta k + \frac{c}{2}(\Delta k)^2. \tag{2.A.3}$$

Subtracting $A\Delta k$ gives equation (2.8); taking expectations conditional on s gives equation (2.9).

2.A.4 Proof of Proposition 2.2

For $k_\theta > 0$, $Ak_\theta < RD_\theta$ is equivalent to $A < R\lambda c k_\theta / 2 = a_\theta$. Substitution gives equation (2.11). With Φ denoting the standard normal distribution function, the shortfall probability $\Phi((a_\theta - \mu)/\sqrt{v})$ increases with a_θ. □

2.B Data and empirical checks

2.B.1 Earnings forecasts

Figure 2.2 uses fixed, disjoint analyst groups within each firm-target. Each group mean uses its members' latest available forecasts issued within 180 days; contributing analysts can change. Revisions are twelve-month log changes, with subsequent revisions ending approximately 3, 15 or 27 months before fiscal year-end. Group means must be positive, but realised earnings are unrestricted. Both group orientations receive equal observation weights. Within each horizon, I trim at the pooled 1st/99th percentiles of either revision before removing fixed effects.

The 1,256 retained observations yield a slope of −0.197 (s.e. 0.099, $p = 0.068$) with year and horizon effects, compared with −0.195 (s.e. 0.082, $p = 0.032$) with horizon effects alone. Standard errors cluster by firm and year; p-values use t_{14} for fifteen year clusters.

2.B.2 Supporting empirics for poduction and debt

Figure 2.3 sums Continental, PDC, SM Energy, Chesapeake, Diamondback and Sanchez; Figure 2.1 uses Continental and SM. Capex includes gas investment and some acquisitions, whereas production covers crude oil and condensate. Company membership is fixed, but asset ownership changes. Quarterly debt includes current maturities and excludes separately reported operating leases; Sanchez's 2019Q3 balance includes bankruptcy reclassifications and financing. WTI averages monthly Cushing prices. The 2014Q2–2015Q4 spending–production divergence survives omitting any one firm and using equally weighted indices.

The aggregate capex and debt changes use Compustat extraction firms (SIC 1311) reporting in US dollars, with varying membership and debt excluding current maturities. Among 160 producers ranked by 2013 leverage, FISD records 7/53 bankruptcies in the bottom third and 14/54 in the top third over 2015–2020; coverage is limited to public-bond issuers.

CoreWeave capex sums four quarters of cash property and equipment purchases; debt excludes current maturities and leases. The revised 2025 cash-capex comparator gives June 2026 spending of $21.456 billion; later filings repeat the older comparator and imply $20.566 billion. This discrepancy remains unresolved.

2.B.3 Oil prices and investment

For the six producers in Figure 2.3, I estimate a VAR in $100\Delta \log X_t$ for WTI, capex and oil production, with one or two quarterly lags and quarter-of-year effects. Both specifications use 26 quarters, 2013Q2–2019Q3.

Table 2.1: Granger tests

	One quarterly lag			Two quarterly lags		
Direction	F	HAC–Holm	HC3–Holm	F	HAC–Holm	HC3–Holm
WTI → Capex	**< 0.001**	**0.038**	0.231	**0.002**	**0.024**	0.971
Capex → Oil production	**0.037**	**0.003**	0.138	0.129	0.240	0.971
Residual degrees of freedom		19			16	

Notes: Joint-exclusion p-values: conventional F and robust HAC (four lags) or HC3 tests. Robust p-values use Holm corrections across six directions per specification. Bold: $p < 0.05$. Sources: company filings and EIA.

The one-lag WTI–capex coefficient is 0.62. Inference weakens under HC3 and some company omissions; neither lag specification rejects for equally weighted company indices. The one-lag production equation has residual serial correlation. These tests measure predictive content, not causal effects.

AI Investment and the Economic Life of Chips

This proposal presents a model I wrote for an idea I am developing to present to Ricardo Reis, as part of our ongoing research on how to model capital depreciation in the age of AI.

3.1 Introduction

How does stronger demand for AI computing affect investment in graphics processing units (GPUs) and the economic life of those already installed? Firms can meet additional demand by buying new equipment or putting existing equipment to work, while constraints on deliveries can raise the earnings of the chips already available. An older chip may therefore become less capable relative to the frontier while earning more for its owner, because it can still serve smaller models and less demanding tasks. For example, I show in preliminary rental data that the H100's hourly rent rises by about 56%, from $1.64 to $2.55, between February and September 2026, whilst its rent relative to the newer B200 falls from 58% to 34%. Understanding economic depreciation therefore requires separating changes in what a chip can do from changes in how much that work pays.

I adapt elements of the putty-putty and putty-clay benchmarks discussed by Atkeson and Kehoe (1999), together with the decomposition of computer depreciation in Geske et al. (2007). I retain a common return to standard compute and the operating decision for equipment with fixed input requirements. With fixed compute capacities, a common return makes gross rents move proportionally across vintages, leaving differences in their exposure to demand unexplained. Geske et al. distinguish replacement-price changes from obsolescence, but do not model the joint determination of rents and investment. I extend these ideas by allowing chips to earn an additional premium from demanding work, according to capabilities that can decline as tasks evolve while the hardware remains functional. Owners value expected net earnings and choose whether to operate, while the supply of new chips determines how stronger demand divides between additional investment and higher asset values.

The model yields four results connecting new investment to the earnings and values of existing chips. When supply slopes upwards, stronger demand raises investment and chip values together, whereas a loss of production capacity raises values while reducing investment. Higher expected frontier premiums raise the value of new chips, but their implications for older equipment depend on how much of the premium that equipment can still earn. More capable chips have greater exposure to these earnings, so a rise in the premium can raise an older chip's dollar rent while reducing its rent relative to a newer chip.

I pair the results with preliminary rental prices, shipment estimates and matched provider quotes. The rental comparisons show that earnings do not move proportionally across generations: newer chips benefit more in the recovery, while older equipment can earn more despite losing ground in relative terms. This motivates allowing chips to differ in their exposure to demanding work, although rents alone neither identify the underlying shocks nor establish how earnings changes affect asset values. The proposed measurements of capabilities, asset prices and operating costs would help separate changing market returns from capability loss.

3.2 The model

I develop a vintage-capital model to examine how the earnings available from different computing tasks affect the operation and value of existing chips, and the incentive to invest in new equipment.

3.2.1 Earnings and the operating decision

I first distinguish the income a chip earns from standard computing from the additional premium it can earn on more demanding tasks, before considering when these earnings cover its operating costs. The benchmark retains the common-service feature of the putty-putty model: old and new equipment provide the same service once measured in efficiency units, and earn a common return per unit (Atkeson and Kehoe, 1999). Time is discrete, and a chip of vintage v, entering service in period v, provides $z_v > 0$ units of standard compute per period, fixed when it is built. The chip fails at the end of each period with probability $\delta \in [0, 1)$, independently of use and with no recovery value.

The extension in this proposal is to allow chips to differ in their access to more demanding work, so that their earnings depend jointly on their capabilities and the market premium for those tasks. This builds on the distinction between physical deterioration and obsolescence in Geske et al. (2007), where equipment can retain its original functionality while becoming less useful as computing requirements change. Standard work, such as serving smaller models, pays $\rho_t > 0$ per unit, while more demanding tasks, such as training larger models, pay an additional premium $s_t \geq 0$, of which the chip can earn a share $\kappa_{v,t} \in [0, 1]$. Its gross rental earnings and the evolution of this capability share are

$$y_{v,t} = z_v(\rho_t + \kappa_{v,t} s_t), \qquad \kappa_{v,t+j} = \kappa_{v,t}(1-d)^j, \quad 0 \leq d < 1. \tag{3.1}$$

New frontier chips enter with $\kappa_{t,t} = 1$, whereas an older chip may earn only part of the premium because some demanding tasks have moved beyond its capabilities. The common decay rate d describes this gradual loss of earning opportunities as tasks evolve, separately from the physical failure captured by δ. The capability share is consequently an economic object to be measured using suitable workloads, rather than an observed benchmark score, and the analysis does not treat d as a rate already recovered from rental prices.

The owner operates the chip when its earnings cover the running cost $\omega_t \geq 0$, which includes the relevant power and hosting expenses and is incurred only while the chip is in use. Because operating today does not change its future condition, the decision rule is $y_{v,t} \geq \omega_t$ and net earnings are $R_{v,t} = \max\{y_{v,t} - \omega_t, 0\}$. I initially hold running costs common across chips, although allowing a cost $e_v w_t$, with fixed energy and space requirements e_v and input price w_t, accommodates differences in efficiency under the same decision rule.

Available computing capacity depends on investment, survival and owners' operating decisions. Letting $N_{v,t}$ denote surviving chips and $u_{v,t}$ the share operating, compute in use is $C_t = \sum_{v \leq t} u_{v,t} z_v N_{v,t}$. Investment adds a new vintage and failure or retirement removes older chips, but firms can also provide more compute by increasing the operating share of equipment already installed. Standard and frontier returns depend on demand and this available capacity. The propositions impose the local restrictions needed for each comparison, while a quantitative equilibrium analysis will require explicit demand functions for the two types of work.

3.2.2 Values and new investment

The decision to retain or purchase a chip depends on earnings over its remaining life, so I next connect expected net earnings to asset values and the supply of new equipment. I assume a risk-neutral owner who discounts at $r > 0$ and can retire the chip at the start of a period for $S_{v,t} \geq 0$. Retaining it yields net rent at the end of the period and, if it survives, its value at the start of the next, so

$$P_{v,t} = \max\left\{S_{v,t},\ \frac{R_{v,t} + (1-\delta)\mathbb{E}_t P_{v,t+1}}{1+r}\right\}. \tag{3.2}$$

The continuation expectation is conditional on survival, and a no-bubble condition rules out value unsupported by future earnings or retirement proceeds. A chip whose current earnings fall below its running cost may nevertheless be worth keeping idle if the owner expects to operate it later, so the model distinguishes temporary idleness from retirement. For a retained chip with positive value, rearranging (3.2) gives a user-cost relation in the spirit of Hall and Jorgenson (1967):

$$R_{v,t} = P_{v,t}\left(r + 1 - \frac{(1-\delta)\mathbb{E}_t P_{v,t+1}}{P_{v,t}}\right). \tag{3.3}$$

The owner must earn enough net rent to cover interest and expected value depreciation, which includes both the possibility of failure and the expected change in the value of a surviving chip. Expected depreciation can be negative if expected appreciation is sufficiently strong, even though the chip continues to face physical failure and may lose capability relative to future tasks. Observed rental earnings, physical deterioration and asset-value depreciation therefore need not move together.

To separate the sources of asset value, I use an always-active benchmark in which the chip operates throughout its surviving life and its retirement option never binds. Iterating (3.2) under these conditions gives

$$P_{v,t} = z_v(A_t + \kappa_{v,t} b_t) - \Omega_t. \tag{3.4}$$

Here A_t is the discounted stream of standard returns, b_t is the discounted premium for a currently fully capable chip, and Ω_t is discounted running cost. Future premium earnings receive the additional capability discount $(1-d)^j$, so the owner values them according to both how long the work is expected to pay a premium and how much of that premium the chip will remain able to earn. Appendix 3.A.1 defines the three components and derives the expression. The benchmark makes this decomposition explicit, but when idling or retirement becomes relevant, valuation must retain the corresponding decisions in (3.2).

Investment takes place when the value buyers attach to a new chip meets the price at which producers can deliver it. Write $c_t > 0$ for baseline acquisition cost before any delivery premium and define $q_t = P_{t,t}/c_t$, following Tobin (1969). Producers deliver I_t chips at a marginal price $c_t \Phi(I_t/\bar{I}_t)$, where $\bar{I}_t > 0$ measures production capacity, $\Phi(0) = 1$ and $\Phi' \geq 0$. The equilibrium investment condition is

$$q_t = \Phi(I_t/\bar{I}_t) \quad \text{if } I_t > 0, \qquad q_t \leq 1 \quad \text{if } I_t = 0. \tag{3.5}$$

The ratio q_t compares asset value with baseline cost, so it can exceed one when producers charge more to deliver additional chips under tight supply. On the demand side, installing more compute lowers prospective earnings and hence the value of a new chip, $\partial P_{t,t}/\partial I_t < 0$, holding expected future investment and other conditions fixed. A rise in demand instead raises chip value at every investment level.

3.3 Propositions and empirical evidence

I derive four propositions linking investment, chip values and rental earnings, and examine their implications using preliminary evidence that also clarifies what further measurements are needed. The evidence uses archived Vast.ai marketplace statistics (Huang, 2026), supplemented by shipment estimates and matched provider quotes. The analytical rental panel contains 1,955 monthly observations for 64 GPU models from October 2022 to September 2026, with prices measured in dollars per GPU-hour before operating costs. I retain model-months averaging at least three rented GPUs and models with at least twelve qualifying months. I use the number of hardware generations behind the newest architecture within each segment (data-centre, workstation or consumer) as a proxy for capability, so the comparisons assess qualitative predictions while asset values, expected returns and κ remain unobserved. I am also in discussions with Silicon Data to obtain daily historical data on GPU rental and purchase prices, term structures, forward curves and residual values, with additional detail across providers and locations. These data would improve our ability to identify the effects of supply and demand shocks by comparing daily rental and asset-price responses across providers and locations with different exposure.

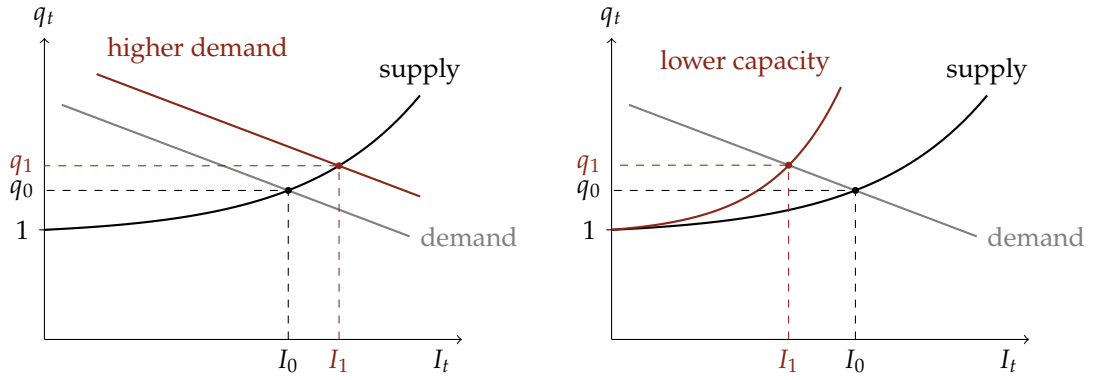
3.3.1 Investment distinguishes demand from supply

I first examine how investment helps distinguish stronger demand from a contraction in supply, since both can raise chip values but have different implications for the quantity installed.

Proposition 3.1 (Investment and scarcity)**.** *Suppose $\partial P_{t,t}/\partial I_t < 0$. At an interior equilibrium with $\Phi' > 0$, a rise in demand that raises chip value at each investment level increases both I_t and q_t. Lower production capacity $\bar{I}_t$ instead reduces investment and raises q_t, holding other determinants fixed. With perfectly elastic supply, higher demand raises investment at $q = 1$, whereas at a binding capacity limit, it raises q at fixed investment.*

Proof. See Appendix 3.A.2.

When demand increases, buyers are willing to pay more for a new chip because they expect it to earn more over its life. They purchase additional chips until the resulting expansion of compute reduces prospective earnings and the higher delivery price absorbs the remaining gain, moving the market along the supply curve in Figure 3.1(a). A production interruption has a different implication: by making chips harder to obtain at every investment level, it leads buyers to install fewer chips while raising the value of the capacity that remains available, as in panel (b).



(a) A rise in demand (b) A fall in capacity

Figure 3.1: Demand and supply have different implications for investment and q

Notes: The demand curve is the value of a new chip relative to baseline cost, $P_{t,t}/c_t$, while the supply curve is $\Phi(I_t/\bar{I}_t)$. Higher demand raises investment and q_t together. Lower production capacity raises q_t while reducing investment. Initial capacity is normalised to one.

Empirical evidence. Figure 3.2 compares estimated shipments of Blackwell chips (B200 and B300) with B200 hourly rents, asking whether additional supply coincides with higher earnings from installed capacity. From 2026 Q1 to Q2, estimated B200 and B300 shipments rise from 1.34 to 1.56 million chips, or 16.2%, while average B200 rent rises from \$3.09 to \$4.50 per GPU-hour, or 45.4%. The joint increase is consistent with a demand expansion that raises rents even as producers deliver more chips.

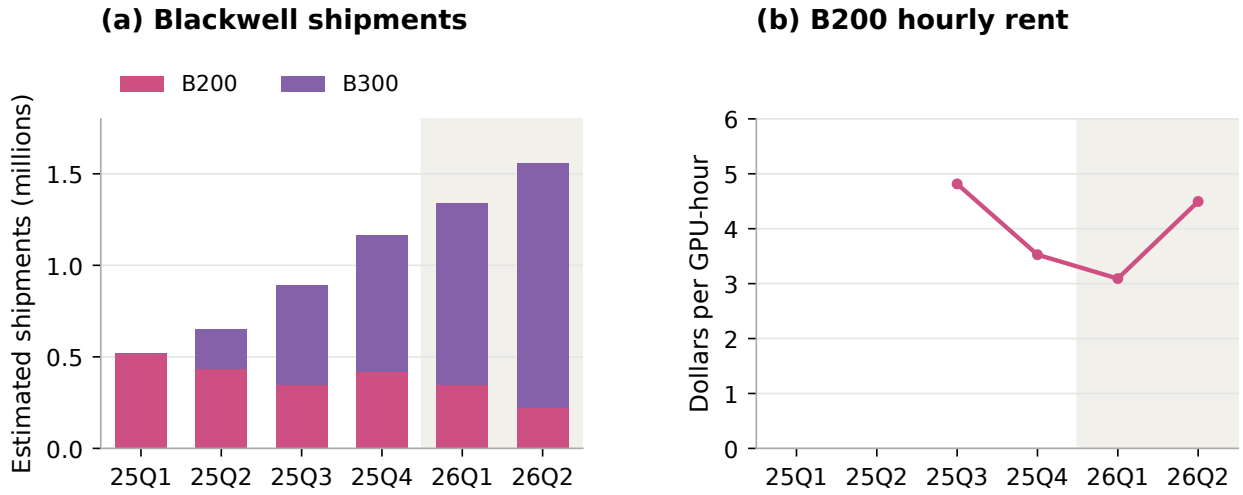


Figure 3.2: Shipments and rental earnings rise together in early 2026

Notes: Sources: archived Epoch AI chip-count estimates and the Vast rental panel. Panel (a) stacks median estimated B200 and B300 shipments, with units inferred from revenues and assumed prices. Panel (b) averages monthly B200 rents within complete quarters. Quarters with fewer than three observations are blank, and incomplete 2026 Q3 shipment estimates are excluded. Product mix shifts towards B300, so the two panels cover different hardware mixes. Shading marks 2026.

Distinguishing this interpretation from other explanations requires measurements closer to the model. Shipment estimates infer units from revenues and assumed prices, deliveries need not coincide with installations, and the shift towards B300 means that B200 rents cover only part of the hardware mix. Rent measures a flow of earnings, whereas q_t compares a chip's asset value with baseline acquisition cost. Matching purchase prices and installations for the same hardware and delivery period would bring the measurements closer to these objects, but recovering q_t also requires an independent measure of baseline cost, and distinguishing shocks requires a credible source of demand or supply variation.

3.3.2 Expected frontier earnings raise new-chip values

The investment response depends on what buyers expect new equipment to earn, so I next examine how expected frontier premiums contribute to a new chip's value as its capability declines.

Proposition 3.2 (The frontier premium and q)**.** *In the always-active benchmark, a new chip satisfies $q_t c_t = z_t(A_t + b_t) - \Omega_t$. Holding standard returns, running costs, compute, baseline cost and capability decay fixed, a higher expected frontier premium at any future date raises q_t.*

Proof. See Appendix 3.A.3.

A high premium today contributes more to a new chip's value when buyers expect that premium to persist and expect the chip to continue earning it, which is why both the expected path of s_t and capability decay enter b_t. With other supply determinants fixed, the higher valuation induces investment through (3.5), although perfectly elastic supply keeps equilibrium q_t at one as installations expand. In that case the frontier premium still contributes to earnings, but discounted expected net earnings just cover baseline acquisition cost.

Empirical evidence. The rental paths in Figure 3.3 show how the earnings available near the frontier pull away from those of older hardware during 2026. Between December 2025 and September 2026, the frontier–legacy gap widens by approximately 44 log points. Since both indices equal 100 in December 2024, the gap does not measure the level of the frontier premium, and changes in standard returns and group composition also enter the comparison. The figure therefore motivates examining which expected earnings buyers capitalise into new-chip values, rather than separately recovering s_t from the indices.

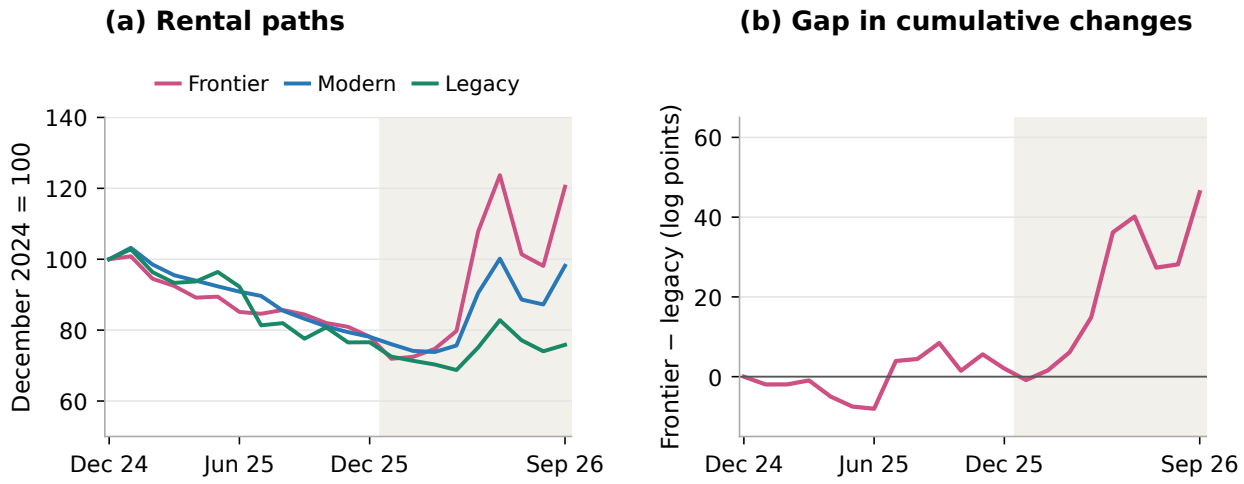


Figure 3.3: Frontier rents pull away from legacy rents in 2026

Notes: Source: Vast tier indices, chaining within-model monthly changes and excluding each model's first three observed months. December 2024 equals 100. Frontier is the newest architecture in each segment, legacy is Turing and older, and modern is the remaining group. Membership can change. Panel (b) is 100 log(frontier index/legacy index). The gap measures different cumulative changes, not the level of a frontier premium. Shading marks 2026.

Matched posted quotes suggest that the recovery extends beyond Vast. After matching hardware, provider, region and recorded server configuration, specialised cloud quotes rise by about 30–32% for A100, H100 and H200 and by 44% for B200 between February and September 2026, based on three, four, three and two providers respectively. These quotes provide an additional observation of improving rental opportunities across venues, although posted prices need not equal realised payments. Testing the valuation result requires connecting those opportunities to acquisition prices and expectations about future rents, since a temporary increase in current earnings need not produce the same value response as one expected to persist.

3.3.3 More capable chips have greater premium exposure

Chips differ in how much of the frontier premium they can earn, raising the question of how the same change in that premium affects earnings and values across vintages.

Proposition 3.3 (Exposure across vintages)**.** *In the always-active benchmark, $\partial P_{v,t}/\partial b_t = z_v \kappa_{v,t}$. At fixed compute, proportional value exposure rises with capability if $z_v A_t > \Omega_t$. At $\rho_t > 0$, percentage rental responses to a premium increase also rise with capability, holding ρ_t and capabilities fixed. The ordering applies to both premium gains and losses.*

Proof. See Appendix 3.A.4.

Because standard work provides a base of earnings, a more capable chip obtains a larger fraction of its rent from the premium and consequently experiences a larger percentage response when that premium changes: $\partial \log y_{v,t}/\partial s_t = \kappa_{v,t}/(\rho_t + \kappa_{v,t} s_t)$ increases with $\kappa_{v,t}$. The same reasoning applies to gains and losses, since greater reliance on these earnings increases exposure in either direction. For asset values, the comparison also depends on running costs. The condition $z_v A_t > \Omega_t$ ensures that standard earnings alone support positive value, so greater capability raises proportional exposure to the premium at fixed compute.

Empirical evidence. I examine this ordering by comparing rental changes across generation groups, retaining models observed at both endpoints of each window and fixing their group at the start. Between December 2025 and September 2026, mean gains are 43.3 log points at the frontier, 20.3 one generation behind and 11.7 further behind, so the recovery is larger for models closer to the frontier (Figure 3.4). The corresponding declines in 2025 range from 27.3 to 35.9 log points and show much less separation. Because the models entering the two windows differ, each panel provides a comparison within its own sample, rather than tracing the same group through a fall and recovery.

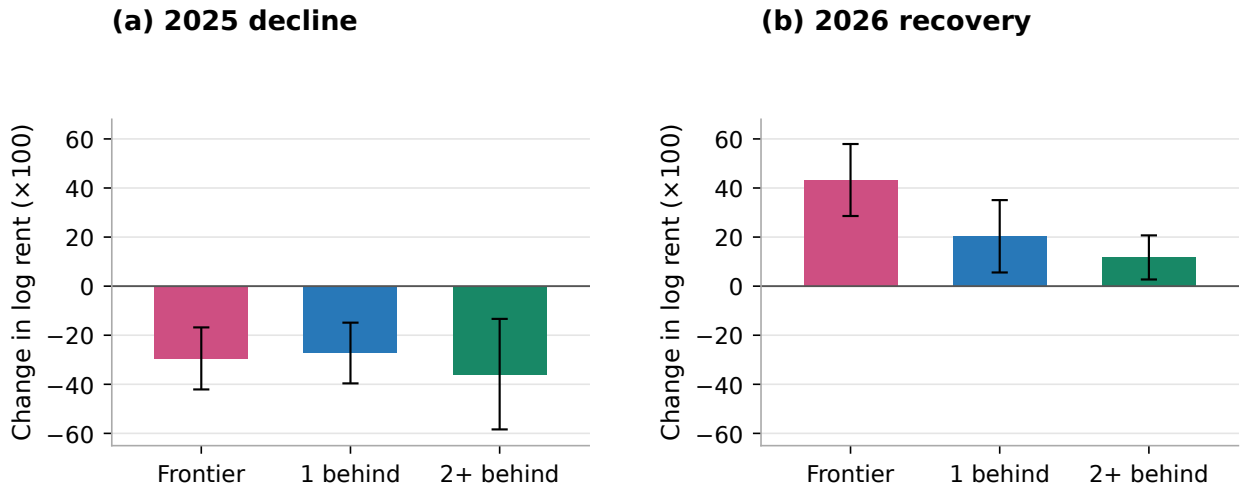


Figure 3.4: Rental gains in 2026 are larger near the frontier

Notes: Source: Vast endpoint estimates. Groups fix generation distance within each segment at the window's start, retaining models present at both endpoints. Bars show mean $100 \log(y_{\text{end}}/y_{\text{start}})$ and 95% heteroskedasticity-robust confidence intervals (HC1). From the frontier back, sample sizes are 14, 17 and 10 models in 2025 and 8, 17 and 32 in 2026. Windows run December 2024–December 2025 and December 2025–September 2026. These groups differ from the changing tiers in Figure 3.3.

The recovery follows the ordering predicted for a premium increase, but the less differentiated 2025 decline gives little support to the corresponding downside ordering. Standard returns, available capacity and demand for particular configurations can also move, so the two episodes may reflect different combinations of shocks. Testing the symmetry in the proposition therefore requires separating changes in standard earnings from changes in the frontier premium, using independent information about capabilities and the work for which demand changes.

3.3.4 An older chip can earn more while falling behind

Greater exposure near the frontier also changes the interpretation of relative rents: an older chip can benefit from stronger demand for demanding work while gaining less, proportionally, than newer equipment.

Proposition 3.4 (Absolute and relative earnings). *For an older–newer pair (o, n) with $0 < \kappa_o < \kappa_n$, a rise in the frontier premium raises the older chip's own rent but reduces its rent relative to the newer chip, holding standard returns, compute and capabilities fixed. With zero running costs and $A_t > 0$, a rise in b_t has the same effects on asset values.*

Proof. See Appendix 3.A.5.

Provided the older chip can still earn some of the premium, it also gains, but the newer chip gains proportionally more because the premium accounts for a larger fraction of its rent. For the fixed pair, relative earnings are

$$\frac{y_{o,t}}{y_{n,t}} = \frac{z_o}{z_n} \frac{\rho_t + \kappa_o s_t}{\rho_t + \kappa_n s_t} \tag{3.6}$$

so a rise in s_t lowers the ratio even as both chips earn more, showing why a decline in relative rent does not establish that older equipment is becoming less profitable to operate.

Empirical evidence. Figure 3.5 illustrates this distinction following the older A100 SXM4 and newer B200 models throughout February–September 2026. The A100's rent rises by 15.5%, from \$0.83 to \$0.96 per hour, while the B200's rent rises from \$2.83 to \$7.50, reducing the A100/B200 ratio from 29.5% to 12.8%. Following a fixed pair rules out a change in the reference model as the explanation for the falling ratio, although the services and configurations surrounding each chip may still change.

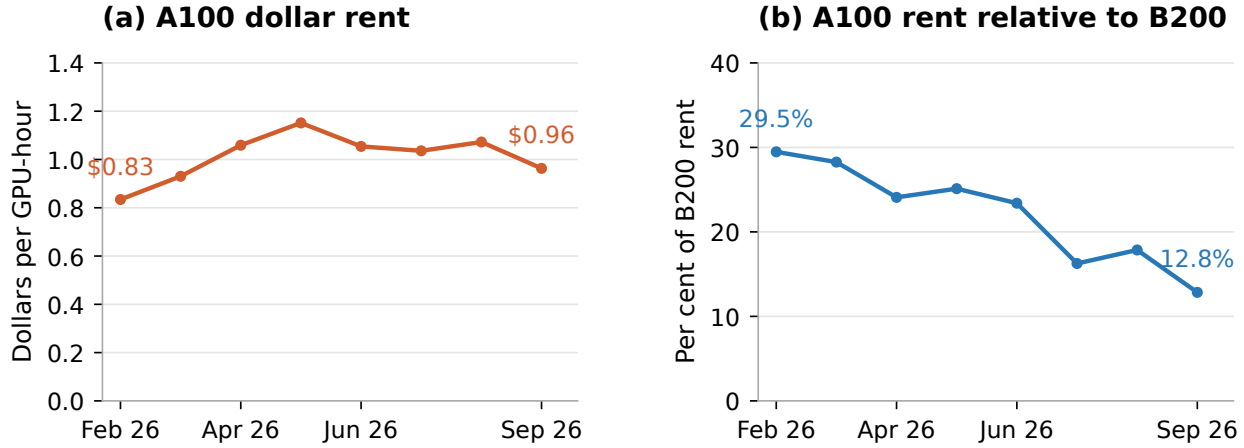


Figure 3.5: The A100 earns more while its relative rent falls

Notes: Source: archived Vast monthly rents for A100 SXM4 and B200, February–September 2026. Panel (a) gives the A100's dollar rent, while panel (b) divides that rent by the B200's rent in the same month. Endpoints are rounded. Both panels hold chip identities fixed, although surrounding service configurations may still change.

The observed price pattern illustrates the proposition without identifying which changes in market returns produced it. In particular, interpreting the falling ratio as capability decay would confound a change in what the chip can earn with a change in the returns to that capability. If both chips lose capability proportionally while task returns remain fixed, the newer chip loses a larger fraction of its total earnings and the older chip's relative rent can instead rise, as Appendix 3.A.5 shows. Independent measures of capability are therefore needed to distinguish capability loss from the market-return changes that can raise dollar earnings while reducing relative earnings.

3.4 Conclusion and next steps

The model connects investment in new chips to the earning opportunities of existing equipment, while the preliminary evidence motivates further measurement to distinguish changing market returns from declining capability. When supply slopes upwards, stronger demand raises investment and asset values together, whereas a production contraction raises values while reducing investment. Expected frontier earnings enter chip values according to remaining capability, while an increase in the current premium can raise an older chip's rent even as its rent falls relative to newer equipment. The preliminary evidence is strongest for these rental patterns, with a larger recovery near the frontier and rising A100 dollar rents despite falling relative rents, while I still need to test the investment and asset-value responses directly.

To distinguish capability decay from changing market returns, I need to connect rental earnings to independent measures of capabilities and asset values. Standard-compute benchmarks can discipline z_v, while repeated tests on demanding workloads can inform $\kappa_{v,t}$ and help separate capability decay from changes in the premium. I would also match acquisition or transaction prices with contract rents, energy requirements and hosting costs to test the user-cost equation for comparable hardware and service configurations. By following individual owners, I could better distinguish a chip leaving one marketplace from becoming idle or being retired, while the cost data would establish whether standard work remains profitable as frontier earnings decline.

To test the investment result, I still need a credible way to distinguish changes in demand from changes in supply. A production disruption would be useful if it demonstrably changes deliveries and can be separated from concurrent demand changes, while exposure to independently measured workload growth could help distinguish standard returns from the frontier premium. The current evidence does not yet provide this design. I would match installations with purchase prices and an independent estimate of baseline costs to measure investment and q_t for the same hardware and delivery period. With demand, supply capacity and operating costs disciplined by data, I could then use the model to quantify how stronger AI demand divides between new investment, greater use of existing equipment and higher rents, and how much investment expands capacity rather than replacing chips whose expected earnings no longer justify retaining them.

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3.A Valuation and proposition proofs

3.A.1 The present-value representation

Suppose the chip operates at every future date conditional on survival, never exercises the retirement option, and discounted expected rents converge. Define $\beta = (1-\delta)/(1+r)$ and $\beta_d = \beta(1-d)$. Iteration of (3.2), with the terminal term vanishing under the no-bubble condition, yields

$$\begin{aligned} P_{v,t} &= \frac{1}{1+r}\sum_{j=0}^{\infty}\beta^j\mathbb{E}_t\left[z_v\rho_{t+j} + z_v\kappa_{v,t}(1-d)^j s_{t+j} - \omega_{t+j}\right] \\ &= z_v(A_t + \kappa_{v,t}b_t) - \Omega_t, \end{aligned} \tag{3.A.1}$$

$$A_t = \frac{1}{1+r}\sum_{j\geq 0}\beta^j\mathbb{E}_t\rho_{t+j}, \quad b_t = \frac{1}{1+r}\sum_{j\geq 0}\beta_d^j\mathbb{E}_t s_{t+j}, \quad \Omega_t = \frac{1}{1+r}\sum_{j\geq 0}\beta^j\mathbb{E}_t\omega_{t+j}. \tag{3.A.2}$$

The multiplicative survival term uses the model's independent failure assumption. When idling or retirement can bind, the maximum in (3.2) must be retained and this linear decomposition need not apply. For any retained chip, including an idle one, multiplying its continuation value by $1+r$ and rearranging nevertheless gives (3.3).

3.A.2 Proof of Proposition 3.1

At positive investment, the downward-sloping valuation curve $P_{t,t}/c_t$ meets the supply curve $\Phi(I_t/\bar{I}_t)$. A rise in demand raises chip value at each investment level, holding baseline cost and other determinants fixed, so value exceeds the supply price at the old quantity. The new intersection must therefore lie further along the upward-sloping supply curve, with both investment and q_t higher.

A fall in production capacity raises the supply price at every positive investment level because

$$\frac{\partial\Phi(I_t/\bar{I}_t)}{\partial\bar{I}_t} = -\Phi'(I_t/\bar{I}_t)\frac{I_t}{\bar{I}_t^2} < 0. \tag{3.A.3}$$

Holding the valuation curve fixed, the new intersection has lower investment and higher q_t. Along a fixed supply curve, the local investment response is

$$d\log I_t = \eta\, d\log q_t, \qquad \eta = \frac{\Phi(I_t/\bar{I}_t)}{(I_t/\bar{I}_t)\Phi'(I_t/\bar{I}_t)} > 0. \tag{3.A.4}$$

With $\Phi \equiv 1$, positive investment pins q_t at one and higher demand raises investment alone. Under a binding hard capacity limit $I_t = \bar{I}_t$, higher demand instead raises $P_{t,t}/c_t$ at fixed investment. These limiting cases apply while the relevant regime continues to hold. If the value of the first chip is below c_t, investment is zero, so small changes need not move the economy away from that corner. □

3.A.3 Proof of Proposition 3.2

For a new chip, $\kappa_{t,t} = 1$. Substitution in (3.A.1) gives $q_t c_t = z_t(A_t + b_t) - \Omega_t$. Holding the other components and c_t fixed,

$$\frac{\partial q_t}{\partial \mathbb{E}_t s_{t+j}} = \frac{z_t \beta_d^j}{(1+r)c_t} > 0. \tag{3.A.5}$$

The weights decline with the horizon through discounting, survival and capability loss. The comparison holds other expected paths fixed, which equilibrium investment can change, so it does not imply that q_t must rise when supply is perfectly elastic. □

3.A.4 Proof of Proposition 3.3

Write $P = z(A + \kappa b) - \Omega > 0$. The level response is $\partial P/\partial b = z\kappa$. The proportional response and its derivative with respect to capability are

$$\frac{\partial \log P}{\partial b} = \frac{z\kappa}{P}, \qquad \frac{\partial}{\partial \kappa}\left(\frac{z\kappa}{P}\right) = \frac{z(zA - \Omega)}{P^2} > 0 \quad \text{if } zA > \Omega. \tag{3.A.6}$$

This comparison holds z fixed. Across chips with different compute and costs, the value ordering needs the corresponding conditions on their standard net value, so capability alone is insufficient. A premium expected at date $t + j$ has level effect $z\kappa\beta_d^j/(1 + r)$.

For current gross rent, $y = z(\rho + \kappa s)$ gives

$$\frac{\partial \log y}{\partial s} = \frac{\kappa}{\rho + \kappa s}, \qquad \frac{\partial}{\partial \kappa}\left(\frac{\kappa}{\rho + \kappa s}\right) = \frac{\rho}{(\rho + \kappa s)^2} > 0. \tag{3.A.7}$$

Compute cancels from this proportional rental response, and integrating the derivative over an increase or decrease in s preserves the ordering when ρ and capabilities remain fixed. □

3.A.5 Proof of Proposition 3.4

For the fixed pair, $\partial y_o/\partial s = z_o\kappa_o > 0$. Differentiating the logarithm of (3.6) gives

$$\frac{\partial \log(y_o/y_n)}{\partial s} = \frac{\rho(\kappa_o - \kappa_n)}{(\rho + \kappa_o s)(\rho + \kappa_n s)} < 0. \tag{3.A.8}$$

With $\Omega = 0$, substitute A and b for ρ and s to obtain the value result. The restriction on running costs ensures that the simple gross-earnings comparison carries over to net asset values. □

For comparison, hold returns constant and let both chips' capability shares be multiplied by $a = (1 - d)^j$. Then

$$\frac{\partial}{\partial a} \log\left(\frac{\rho + a\kappa_o s}{\rho + a\kappa_n s}\right) = \frac{\rho s(\kappa_o - \kappa_n)}{(\rho + a\kappa_o s)(\rho + a\kappa_n s)} \leq 0. \tag{3.A.9}$$

As a falls, the older chip's relative rent weakly rises because the newer chip loses a larger proportion of its earnings. A fixed-pair relative decline therefore cannot by itself be interpreted as common capability decay, while a comparison with a continually replaced frontier reference measures a different object.